2. The Filesystem Tree: /, /etc, /var, /home, /proc

02. The Filesystem Tree: /, /etc, /var, /home, /proc

Phase: Phase 1 — Shell and filesystem
Estimated time: 1–2 days
Slug: `02-filesystem-tree`

Learning objectives

- Navigate with cd, pwd, ls
- Explain key FHS directories
- Create, move, copy, remove files carefully
- Use tab completion and wildcards

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

Filesystem Hierarchy (FHS) highlights

Path	Purpose
`/`	root of the tree
`/bin`, `/usr/bin`	user commands
`/etc`	configuration
`/home`	user home directories
`/var`	logs, caches, variable data
`/tmp`	temporary (often cleared)
`/proc`, `/sys`	kernel/process info (virtual)
`/dev`	device nodes

Navigation and file ops

```
pwd
cd /etc
ls -la
mkdir -p ~/lab/linux-goodway
cp file.txt file.bak
mv oldname newname
rm file.txt          # careful
rm -r dirname        # careful; prefer trash tools when learning
```

Wildcards

`*` any string, `?` one char, `[abc]` character class.

```
ls *.log
```

Common mistakes

- Running `rm -rf /` or similar disasters
- Treating `/proc` files like normal disk files to edit casually
- Filling `/` by writing huge files to `/tmp` unchecked

Exercises

Work in `exercises/02-filesystem-tree/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/02-filesystem-tree/` until you have an honest failing attempt.

Self-check

1. What lives in `/etc`?
2. What is special about `/proc`?
3. What does `mkdir -p` do?

Further reading

- https://refspecs.linuxfoundation.org/FHS_3.0/fhs/index.html